ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers, sensible project decisions and neat, recognisable sketches, circuits and charts.

QUESTION 1

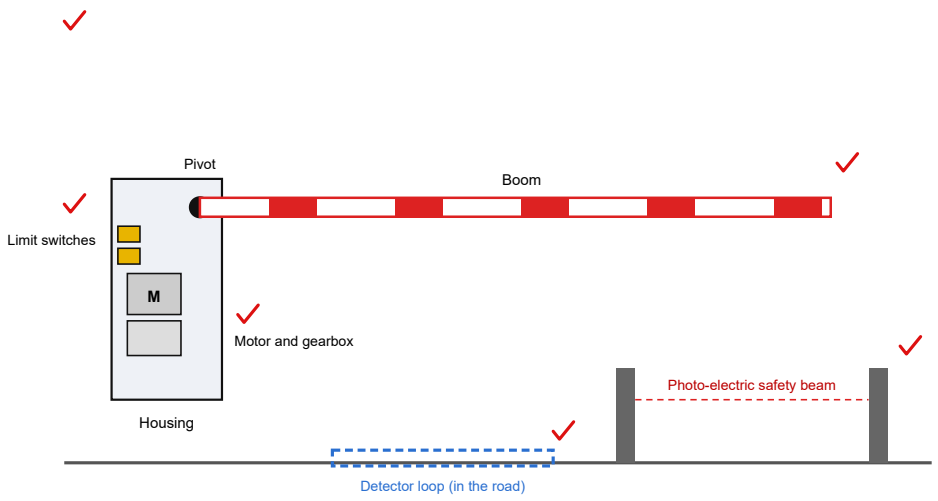
1.1 It checks whether the project is possible and worth doing ✓ – technically, within the budget and time, and safely – before money is spent on it ✓. (2)

1.2 Any FOUR headings ✓ each with a sensible requirement ✓, e.g. Function – lift the boom only for authorised cars; Performance – open in 3 s, at least 300 cycles a day; Safety – never close on a car, emergency stop; Environment – work outdoors in rain and heat (IP65); Cost – within the budget; Maintenance – easy access to the motor and gearbox. (8)

[10]

QUESTION 2

2.1



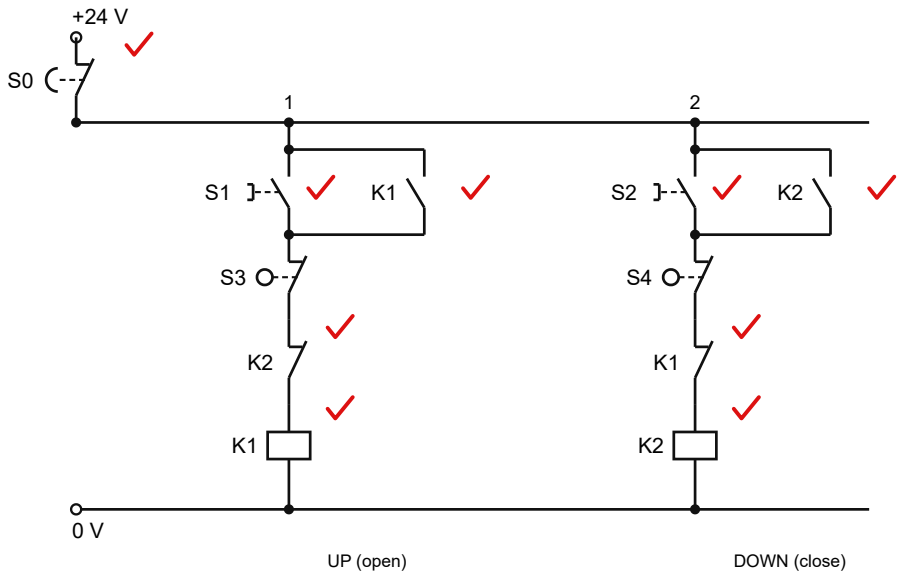
Neat freehand sketch ✓ showing: boom and pivot ✓; housing with motor and gearbox ✓; limit switches ✓; detector loop and safety beam ✓ (5)

2.2.1 $T = F \times d = m \times g \times d$ ✓ $= 12 \times 9{,}81 \times 2$ ✓ $= 235{,}44\ N{\cdot}m$ ✓ (3)

2.2.2 $T_{motor} = T_{boom} \div (GR \times \eta)$ ✓ $= 235{,}44 \div (50 \times 0{,}8)$ ✓ $= 5{,}89\ N{\cdot}m$ ✓ (3)

2.2.3 $90°$ in 3 s $= 0{,}25$ r in 3 s $= 5$ r/min at the boom ✓; motor speed $= 5 \times 50 = 250$ r/min ✓ (2)

2.3



UP push button S1 with K1 holding contact ✓ and DOWN push button S2 with K2 holding contact ✓; common STOP / E-stop ✓; NC K2 contact in the K1 circuit and NC K1 contact in the K2 circuit (interlocks) ✓✓; NC limit switch S3 in the UP circuit ✓ and NC limit switch S4 in the DOWN circuit ✓; correct symbols and labels ✓ (8)

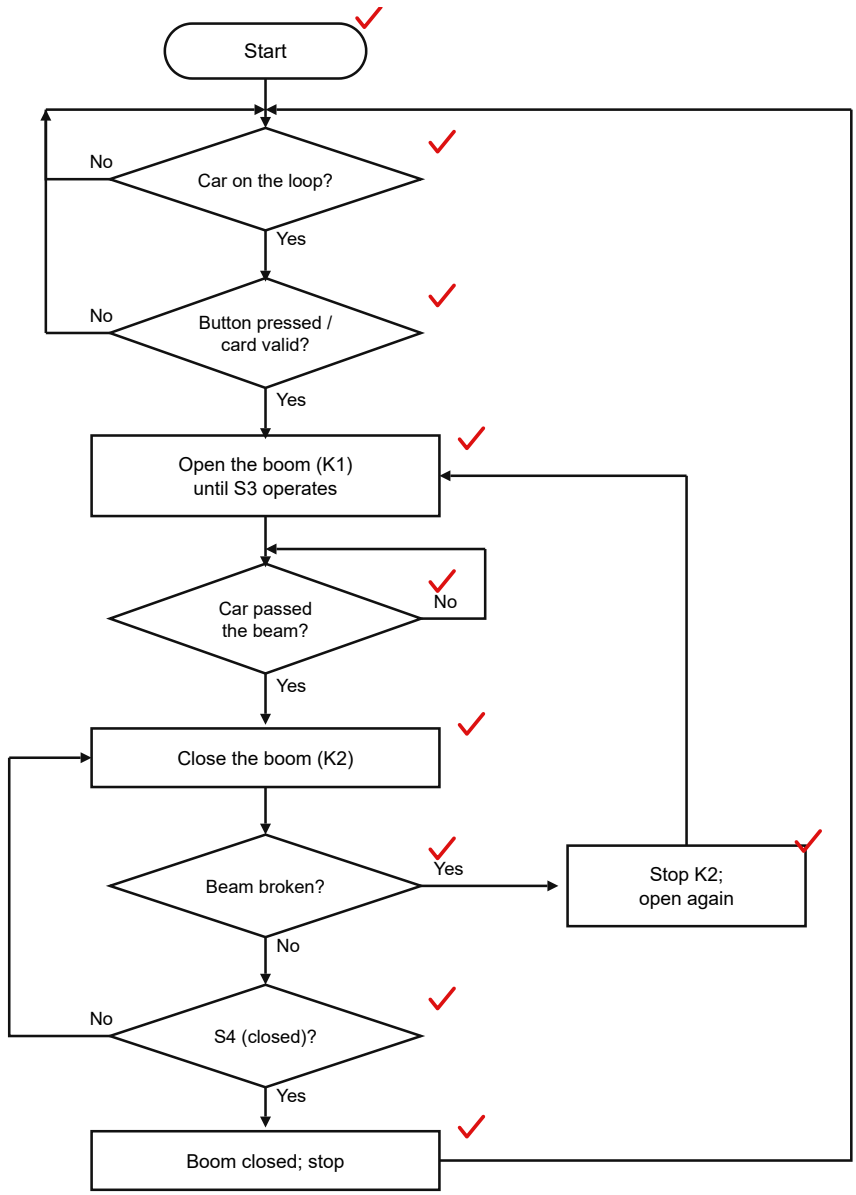
2.4 A car (or person) under the boom would be hit and damaged / injured ✓, so the boom must be lifted clear again until the beam is clear ✓. (2)

2.5 Any TWO ✓✓: inductive detector loop in the road; photo-electric beam; ultrasonic sensor; radar / microwave sensor; camera with number-plate recognition. (2)

[25]

QUESTION 3

3.1



Start ✓; decision: car on the loop? ✓; decision: button pressed / card valid? ✓; open the boom (K1) until S3 ✓; decision: car passed the beam? ✓; close the boom (K2) ✓; decision: beam broken while closing? ✓ – yes: stop and open again ✓; boom closed at S4 ✓; loop back to wait for the next car ✓

(10)

3.2

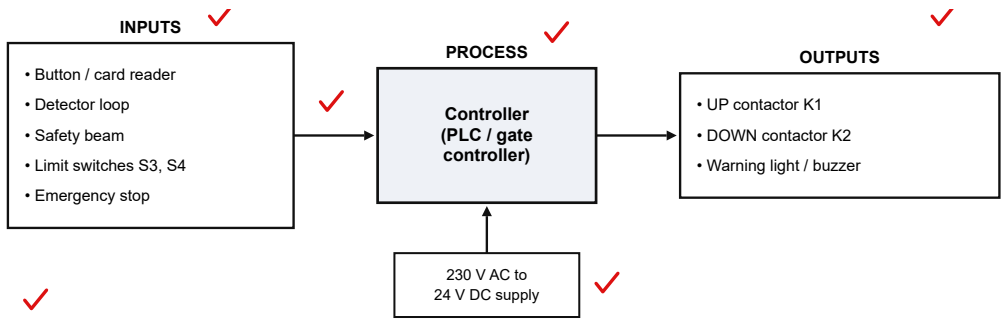
Item	Description	Quantity	Unit cost (R)	Total (R)
1	Boom (aluminium, 4 m)	1	1 800	1 800
2	Gear motor 24 V DC	1	4 500	4 500
3	Steel housing	1	2 200	2 200
4	Limit switches	2	350	700
5	Detector loop and loop detector	1	2 600	2 600
6	Photo-electric safety beam	1	1 200	1 200
7	Controller and contactors	1	3 800	3 800

Suitable headings ✓✓; any SIX sensible items ✓✓✓✓ (2 items per mark); totals and a grand total ✓✓

(8)

3.3 A part drawing shows ONE part with all the dimensions, tolerances and material needed to make it ✓, used in the workshop to manufacture the part ✓. An assembly drawing shows how all the parts fit together, with a parts list ✓, used to assemble and install the gate ✓. (4)

3.4



Inputs ✓ (button / card reader, detector loop, safety beam, limit switches S3 and S4, E-stop); controller ✓; outputs ✓ (UP contactor K1, DOWN contactor K2, warning light / buzzer); power supply ✓; arrows showing the direction of signals ✓; labels ✓

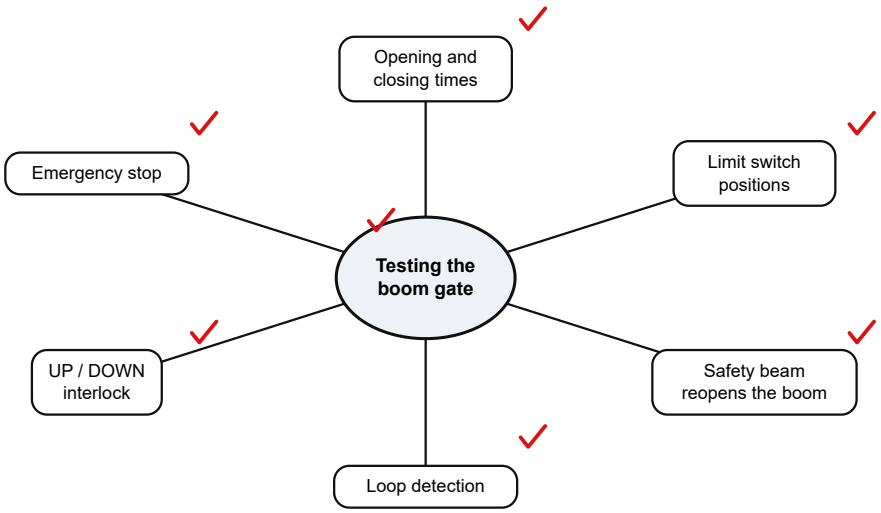
(6)

3.5 So that every task is done in the right order and on time ✓, people, parts and money are available when needed, and delays are seen early ✓. (2)

[30]

QUESTION 4

4.1



Centre topic "Testing the boom gate" ✓ with any FIVE areas ✓✓✓✓✓: opening and closing times; limit switch positions; safety beam reopens the boom; loop detection; interlock (UP and DOWN never together); emergency stop; card reader / button.

(6)

4.2 Move the boom slowly (by hand or jogging) to the fully open position ✓ and adjust the cam / switch S3 so that it just operates there ✓; repeat for S4 at the closed position and test that the motor stops correctly ✓. (3)

4.3 A list of small faults and unfinished work ✓ found during the final inspection, which must be fixed before the project is accepted ✓. (2)

4.4 Any FOUR ✓✓✓✓: normal operation; opening the gate manually during a power failure; what to do when a fault occurs; daily safety checks; resetting the emergency stop; who to contact for service. (4)

[15]

QUESTION 5

5.1

Task	How often
Test the safety beam and loop ✓	Weekly
Check the boom and its fixings ✓	Monthly
Check the limit switch settings ✓	Monthly
Check the gearbox oil and grease the pivot ✓	Every 6 months

Any FOUR sensible tasks ✓✓✓✓ with a sensible frequency for each ✓✓✓✓

(8)

5.2 Any FOUR causes ✓✓✓✓ with a matching check ✓✓✓✓, e.g. safety beam blocked or dirty – clean and test the beam; beam misaligned – check its LED and realign; car still detected on the loop – check the loop detector; faulty K2 contactor or coil – measure the coil voltage and resistance; S4 or S3 wired wrongly / stuck – test continuity; K1 holding contact stuck – check the contactor. (8)

5.3 Low or dirty oil causes wear and overheating of the gears ✓, which leads to breakdown and costly repairs ✓. (2)

5.4 Any TWO ✓✓: faults are reported immediately to a technician; the number of cycles is counted for planned maintenance; the gate can be opened remotely; less travel time for checks. (2)

[20]

TOTAL: 100

END OF MARKING GUIDELINE